FEU INSTITUTE OF TECHNOLOGY

COURSE CODE	COURSE TITLE	UNITS / TYPE
CCS0047	Number Theory	3 UNITS LEC
PREREQUISITE	DISCRETE STRUCTURES 1 (CS0001)	
CO-REQUISITE/S	NONE	
COURSE DESCRIPTION	This course provides a comprehensive study of the fundamental properties of integers and their computational applications. It covers essential foundations ranging from the real number system and mathematical induction to advanced concepts such as divisibility theorems, modular arithmetic, and the theory of congruences. Students will explore prime distribution, primitive roots, and the law of quadratic reciprocity. The course culminates in the application of these mathematical foundations to modern cryptology, specifically the design, encryption, and decryption of secrecy systems such as the RSA cryptosystem.	

INSTITUTION VISION STATEMENT

FEU Institute of Technology aims to be the global standard of innovation in education, research, and nation-building by creating the next generation of changemakers that will shape the future of the country and the world.

INSTITUTION MISSION STATEMENT

FEU Institute of Technology is dedicated to provide quality, relevant, innovative, and industry-based education producing competent and principled professionals with greater sense of responsibility, social awareness and high competitiveness contributing significantly to the betterment of the society.

EDUCATIONAL PHILOSOPHY

FEU Tech's educational philosophy is rooted in transformative learning, emphasizing innovation, relevance, and social responsibility. We aim to cultivate an ecosystem of learning that shapes the next generation of leaders and innovators, driving positive change both locally and globally.

DEPARTMENT VISION STATEMENT

The Computer Science department provides specialized programs in Computer Science whose graduates are internationally recognized as innovative and specialized computing professionals.

DEPARTMENT MISSION STATEMENT

The Computer Science department is committed to provide the students and faculty with an open environment that imbues professional and personal growth through research and development.

PROGRAM EDUCATIONAL OUTCOMES

The graduates of the Bachelor of Science in Computer Science program are:

1. engaged in further professional development and have interest in or aptitude for advanced studies or trainings in computing.
2. entrepreneurs or are employed in computing industries, organizing and managing team-based projects leading to successful and sustainable computing systems solutions.
3. responsible computing professionals actively participating in community groups that make a significant impact in addressing current and future societal challenges.

STUDENT OUTCOMES

a. Apply knowledge of computing appropriate to the discipline.
b. Analyze a complex problem and identify and define the computing requirements appropriate to its solution.
c. Design, implement, and evaluate computer-based systems or applications to meet desired needs and requirements.
d. Function effectively as a member or leader of a development team, recognizing the different roles within a team to accomplish a common goal.

e. Understand professional, ethical, legal, security, and social issues and responsibilities in the utilization of information technology.
f. Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and a clear manner.
g. Analyze the local and global impact of computing information technology on individuals, organizations, and society.
h. Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development.
i. Apply knowledge through the use of current techniques and tools necessary for the IT profession.
j. Apply mathematical foundations, algorithmic principles, and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the tradeoffs involved in design choices and applicable standards.
k. Apply mastery in their field of specialization

GRADUATE ATTRIBUTES

These are the skills and qualities our computing graduates will have when they finish the program.

1. Academic Education
2. Knowledge for Solving Computing Problems
3. Problem Analysis
4. Design/ Development of Solutions
5. Modern Tool Usage
6. Individual and Team Work
7. Communication
8. Computing Professionalism and Society
9. Ethics
10. Life-long Learning

COURSE OUTCOMES AND RELATIONSHIP TO STUDENT OUTCOMES

COURSE LEARNING OUTCOMES (CLO)	ASSESSMENT	STUDENT OUTCOMES											
		a	b	c	d	e	f	g	h	i	j	k	l
1. Prove mathematical propositions and evaluate the structural properties of integers, including divisibility, prime factorization, and algorithmic problem-solving using mathematical induction and the Euclidean algorithms.	SA1, SA2, SA3, MIDTERM EXAM, FINAL EXAM	R	R								R		R
2. Apply modular arithmetic, the theory of congruences, and advanced residue systems to solve complex systems of linear equations and computational problems.	SA1, SA2, SA3, MIDTERM EXAM, FINAL EXAM	R	R								R		R
3. Design and Evaluate secure cryptographic models, specifically public-key systems like RSA, by synthesizing number theoretic principles for data security applications.	SA1, SA2, SA3, MIDTERM EXAM, FINAL EXAM	R	R	I		I					R		R

WEEK/ NO. OF HOURS	INTENDED LEARNING OUTCOME (ILO)	DETAILED COURSE CONTENT	TEACHING AND LEARNING ACTIVITY	ASSESSMENT TASK (AT)	CLO
1 / 1.83 hrs	**Module 0: Course Orientation**				
	• Orientation	• Introduction of the subject • Discussion of Syllabus • Course Requirements • Classroom policies	● Syllabus ● Discuss Course Requirements ● Discussion of ● Classroom Policies		
1-2 / 5.49 hrs	1. Analyze the structural properties of the real number system and the subset of integers. 2. Apply the principle of	**Module 1: Foundations of Discrete Math** **• Subtopic 1: Real Number System and Set Notations** ‣ Types (Sets) of Numbers	• Class Discussion (PPT) • Guided Formative • View Video Courseware	• Formative Assessment 1 (Problem Set) • Seatwork	CLO1

	mathematical induction to prove numerical identities and inequalities. 3. Evaluate the completeness and well-ordering principles as the logical basis for integer arithmetic. 4. Analyze the properties of prime and composite numbers, and apply the Fundamental Theorem of Arithmetic to integer factorization.	‣ Important Notations for Sequences and Series **• Subtopic 2: Well-Ordering Principle and Principle of Mathematical Induction** ‣ Peano Axioms ‣ Successor Operation ‣ The Well-Ordering Principle ‣ Proof by Induction **• Subtopic 3: Prime Numbers** ‣ Primes ‣ Composite ‣ Prime Number Generation ‣ Primality Testing ‣ Fundamental Theorem of Arithmetic	• Supplementary Materials		
3-4 / 7.32 hrs	5. Implement the Division Algorithm to determine quotients and remainders for any integer pair. 6. Execute the Euclidean Algorithm and its extended version to find Greatest Common Divisors (GCD) and Bezout coefficients. 7. Construct solutions for Linear Diophantine Equations and identify the existence of modular inverses.	**Module 2: The Algorithmic Integer** **• Subtopic 1: Divisibility Theorems, Division Algorithm, & Euclidean Algorithm** ‣ Divisibility (The Boolean Operator of Math) ‣ Properties of Divisibility ‣ Greatest Common Divisor (GCD) ‣ Euclid's Lemma ‣ The Euclidean Algorithm **• Subtopic 2: Bézout's Identity & Extended Euclidean Algorithm (EEA)** ‣ Bezout's Identity ‣ The Set of Linear Combinations ‣ The Extended Euclidean Algorithm (Reverse Engineering) **• Subtopic 3: Linear Diophantine Equations & The Modular Inverse** ‣ Linear Diophantine Equations ‣ The Modular Inverse	• Class Discussion (PPT) • Guided Formative • View Video Courseware • Supplementary Materials	• Formative Assessment 2 (Problem Set) • Seatwork • Summative Assessment 1 (M1 to M2)	CLO1
5-6 / 7.32 hrs	8. Perform arithmetic operations within modular systems and simplify complex congruences. 9. Solve systems of linear congruences using the Chinese Remainder Theorem (CRT).	**Module 3: Modular Arithmetic & Systems** **Subtopic 1: Congruence Basics** ‣ Introduction to Congruence ‣ Properties of Congruence ‣ Equivalence Classes **Subtopic 2: Linear Congruences** **Subtopic 3: Chinese Remainder Theorem (CRT)**	• Class Discussion (PPT) • Guided Formative • View Video Courseware • Supplementary Materials	• Formative Assessment 3 (Problem Set) • Seatwork	CLO2
7 / 1.83 hrs	**MIDTERM EXAMINATION**			Midterm Examination (M1-M3)	
7-9 / 9.15 hrs	10. Execute the Square-and-Multiply algorithm to optimize modular exponentiation for large-	**Module 4**: Orders and Primality **• Subtopic 1: Modular Exponentiation**	• Class Discussion (PPT) • Guided Formative	• Formative Assessment 4 (Problem Set) • Seatwork • Summative Assessment 2 (M3	CLO2

	scale computations. 11. Apply Fermat's Little Theorem and Wilson's Theorem to verify the primality of integers. 12. Calculate Euler's Totient Function for various moduli to simplify modular exponentiation. 13. Solve for discrete logarithms using Index Arithmetic and Primitive Roots.	‣ Modular Exponentiation ‣ The Square-and-Multiply Algorithm **• Subtopic 2: Fermat's Little & Wilson's Theorems** **• Subtopic 3: Euler's World** ‣ Euler's Totient Function ‣ Euler's Theorem **• Subtopic 4: Primitive Roots** ‣ What are Primitive Roots? ‣ The Order of an Element ‣ Reduced Residue System **• Subtopic 3: Index Arithmetic** ‣ Properties of the Index ‣ Arithmetic Indices and Congruence	• View Video Courseware • Supplementary Materials	to M4)	
10-12 / 9.15 hrs	14. Evaluate the solvability of quadratic congruences using Euler's Criterion and Legendre symbols. 15. Apply the Law of Quadratic Reciprocity to simplify complex residue calculations. 16. Formulate proofs and solutions for quadratic residues in both prime and composite moduli.	**Module 5: Quadratic Residues** **• Subtopic 1: Introduction to Quadratic Residues** **• Subtopic 2: Euler's Criterion** **• Subtopic 3: Legendre & Jacobi Symbols** ‣ Legendre Symbol - Special Rule for 2 ‣ Legendre Symbol - Special Rule for -1 **• Subtopic 4: Law of Quadratic Reciprocity**	• Class Discussion (PPT) • Guided Formative • View Video Courseware • Supplementary Materials	• Formative Assessment 5 (Problem Set) • Seatwork	CLO2
12-14 / 7.32 hrs	17. Apply classic cryptographic techniques, such as the Caesar and Vigenère ciphers, to encrypt and decrypt basic messages. 18. Evaluate the Diffie-Hellman Key Exchange protocol for secure key distribution over public channels using modular arithmetic. 19. Design and simulate the RSA Cryptosystem for public-key encryption and decryption.	**Module 6: Applied Cryptography** **• Subtopic 1: Classic Era: Caesar Cipher & Vigenère Cipher** ‣ Basic Terminologies ‣ Character Ciphers ‣ Block and Stream Ciphers Vigenere Cipher **• Subtopic 2: Public Key Era: Diffie-Hellman Key Exchange** **• Subtopic 3: The King of Crypto: RSA**	• Class Discussion (PPT) • Guided Formative • View Video Courseware • Supplementary Materials	• Formative Assessment 6 (Problem Set) • Seatwork • Summative Assessment 3 (M5 to M6)	CLO3
14 / 2 hrs	**FINAL EXAMINATION**				
TOTAL HOURS	51.24 hours				

INSTRUCTIONAL MATERIALS

- Lecture Materials (PPT)/ Supplementary and Reinforcement Materials
- Video Materials
- Study Guide

CLO	EXAMS / ASSESSMENTS
1	Summative Assessment 1, Midterm Exam, Final Exam
2	Summative Assessment 2, Midterm Exam, Final Exam
3	Summative Assessment 3, Final Exam

GRADING SYSTEM

Midterm Grade	Final Grade
Midterm Grade (MG) = 60% (Class Standing) + 40% (Midterm Exam) **<u>Lecture: 100%</u>** **Class Standing (CS) 60%** 1. Average of at least three long quizzes 2. Teacher's Evaluation 3. Class participation (Seatwork, Assignments, Recitations) **Midterm Exam (ME) 40%** **PASSING RAW SCORE: 70**	**Final Grade (FG) = 60% (Class Standing) + 15% (Midterm Exam) + 25% (Final Exam)** **<u>Lecture: 100%</u>** **Class Standing (CS) 60%** 1. Average of at least three long quizzes 2. Teacher's Evaluation 3. Class participation (Seatwork, Assignments, Recitations) **Midterm Exam (ME) 15%** **Final Exam (FE) 25%** **PASSING RAW SCORE: 70**

TEXTBOOK/REFERENCES/ONLINE REFERENCES

- Burton, D. M. (2011). Elementary number theory (7th ed.). McGraw-Hill Education.
- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to algorithms (4th ed.). MIT Press.
- Rosen, K. H. (2011). Elementary number theory and its applications (6th ed.). Pearson.
- Silverman, J. H. (2013). A friendly introduction to number theory (4th ed.). Pearson.
- Trappe, W., & Washington, L. C. (2020). Introduction to cryptography with coding theory (3rd ed.). Pearson.

Journals:
Online Journal:
EBSCO
URL: http://search.ebscohost.com

Other Journals (Available in the library):
MIS Quarterly
Journal of the ACM

Online References:
• Courseware Materials available at https://feu.instructure.com
• FEUTECH e-book library collection at
https://drive.google.com/drive/folders/15eccuUWwMrcnbHSNMvqn6HWMMyQB3d2?usp=sharing

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